StudySapiens Board MCQs

Chapter 1 — Biodiversity and Classification

Board-style MCQs · Questions first, answer key with brief reasons at the end · PECTAA 2025–26

Choose the best option. Each question carries one mark. The answer key with brief reasons is given at the end of this chapter.

1. The five-kingdom classification system was proposed by:

(a) Carolus Linnaeus
(b) Ernst Mayr
(c) Carl Woese
(d) Robert Whittaker

2. The three-domain system of classification was proposed in 1990 by:

(a) Theodor Schwann
(b) Robert Whittaker
(c) Louis Pasteur
(d) Carl Woese

3. Which of the following is NOT one of the three domains of life?

(a) Archaea
(b) Monera
(c) Bacteria
(d) Eukarya

4. In the five-kingdom system, all prokaryotes were placed in kingdom:

(a) Fungi
(b) Protista
(c) Monera
(d) Plantae

5. Archaea differ from Bacteria in having membrane lipids with:

(a) Ether linkages and branched chains
(b) Peptidoglycan layers
(c) Cholesterol as the main lipid
(d) Ester linkages and unbranched chains

6. The cell wall of Archaea lacks:

(a) Pseudopeptidoglycan in some species
(b) Polysaccharides
(c) Proteins
(d) Peptidoglycan

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7. Which metabolic process is unique to Archaea?

(a) Fermentation
(b) Photosynthesis
(c) Methanogenesis
(d) Nitrogen fixation

8. Halobacteria are Archaea that live in:

(a) Human intestine only
(b) Extremely saline environments
(c) Hot springs only
(d) Acidic mine drainage only

9. Thermococci are Archaea typically found in:

(a) Hot environments
(b) Animal skin
(c) Freshwater lakes
(d) Frozen soils

10. Methanogens are important in sewage treatment because they:

(a) Cause malaria
(b) Fix atmospheric nitrogen
(c) Carry out anaerobic digestion and produce biogas
(d) Produce oxygen

11. The genetic material of Bacteria is usually:

(a) RNA only
(b) A single circular chromosome
(c) Multiple linear chromosomes with histones
(d) DNA enclosed in a nuclear membrane

12. Bacterial ribosomes are:

(a) 70S
(b) 60S
(c) 80S
(d) 40S

13. Small circular extra-chromosomal DNA molecules in bacteria are called:

(a) Nucleoids
(b) Mesosomes
(c) Capsules
(d) Plasmids

14. Rod-shaped bacteria are called:

(a) Bacilli
(b) Vibrios
(c) Cocci
(d) Spirilla

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15. Comma-shaped bacteria are called:

(a) Spirilla
(b) Cocci
(c) Bacilli
(d) Vibrios

16. Bacteria arranged in chains are called:

(a) Staphylococci
(b) Streptococci
(c) Diplococci
(d) Sarcinae

17. Which structure helps bacteria attach to surfaces and exchange genetic material?

(a) Capsule
(b) Pili
(c) Endospore
(d) Flagella

18. The highest taxonomic rank is:

(a) Class
(b) Kingdom
(c) Phylum
(d) Domain

19. The most specific taxonomic rank is:

(a) Family
(b) Order
(c) Genus
(d) Species

20. In the taxonomic hierarchy, the rank immediately below class is:

(a) Genus
(b) Phylum
(c) Order
(d) Family

21. Humans belong to the order:

(a) Carnivora
(b) Primates
(c) Chiroptera
(d) Rodentia

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22. The family of humans is:

(a) Homininae
(b) Mammalia
(c) Primata
(d) Hominidae

23. In the textbook table, onion is placed in the phylum/division:

(a) Angiosperms
(b) Bryophyta
(c) Gymnosperms
(d) Chordata

24. Organisms of kingdom Fungi have cell walls made of:

(a) Cellulose
(b) Peptidoglycan
(c) Silica
(d) Chitin

25. Fungi obtain nutrition by:

(a) Chemosynthesis
(b) Ingestion
(c) Photosynthesis
(d) Absorption

26. Which group of protists is animal-like and heterotrophic?

(a) Protozoa
(b) Algae
(c) Oomycota
(d) Myxomycota

27. Algae are plant-like protists because they:

(a) Have chlorophyll and are autotrophic
(b) Have chitin walls
(c) Are all multicellular
(d) Ingest food

28. Slime molds and water molds are:

(a) Fungi-like protists
(b) Bacteria
(c) Animals
(d) True fungi

29. Malaria is caused by a member of kingdom Protista named:

(a) Plasmodium
(b) Trypanosoma
(c) Paramecium
(d) Entamoeba

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30. Sleeping sickness is caused by:

(a) Euglena
(b) Trypanosoma
(c) Entamoeba histolytica
(d) Plasmodium

31. The largest group of fungi is:

(a) Ascomycota
(b) Zygomycota
(c) Basidiomycota
(d) Oomycota

32. Rhizopus belongs to:

(a) Basidiomycota
(b) Deuteromycota
(c) Zygomycota
(d) Ascomycota

33. Mushrooms and rusts belong to:

(a) Myxomycota
(b) Ascomycota
(c) Basidiomycota
(d) Zygomycota

34. Penicillin is obtained from:

(a) Penicillium
(b) Agaricus
(c) Rhizopus
(d) Saccharomyces

35. Non-vascular plants are called:

(a) Bryophytes
(b) Angiosperms
(c) Gymnosperms
(d) Pteridophytes

36. Gymnosperms are characterized by:

(a) Seeds enclosed in fruits
(b) Naked seeds
(c) Absence of seeds
(d) Spores only

37. The most diverse plant group is:

(a) Angiosperms
(b) Bryophytes
(c) Pteridophytes
(d) Gymnosperms

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38. Animals are best described as:

(a) Prokaryotic heterotrophs
(b) Multicellular ingestive heterotrophs
(c) Unicellular absorptive heterotrophs
(d) Multicellular autotrophs

39. Sponges belong to phylum:

(a) Cnidaria
(b) Porifera
(c) Platyhelminthes
(d) Mollusca

40. Water enters the body of a sponge through:

(a) Spongocoel
(b) Ostia
(c) Choanocytes
(d) Osculum

41. The collar cells of sponges are called:

(a) Pinacocytes
(b) Cnidocytes
(c) Choanocytes
(d) Amoebocytes

42. Resistant capsules formed by some sponges for survival are:

(a) Gemmules
(b) Spores
(c) Cysts
(d) Buds

43. Cnidarians are:

(a) Diploblastic
(b) Acoelomate
(c) Pseudocoelomate
(d) Triploblastic

44. The stinging organelle of cnidarians is:

(a) Nematocyst
(b) Radula
(c) Flame cell
(d) Choanocyte

45. Which cnidarian body form is free-swimming and reproduces sexually?

(a) Medusa
(b) Polyp
(c) Larva only
(d) Planula

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46. Corals build reefs mainly by secreting:

(a) Silica
(b) Cellulose
(c) Calcium carbonate
(d) Chitin

47. Flatworms belong to phylum:

(a) Annelida
(b) Nematoda
(c) Mollusca
(d) Platyhelminthes

48. Flatworms are acoelomate because:

(a) They have a true coelom
(b) They have a pseudocoelom
(c) They lack a body cavity
(d) They have a hemocoel

49. The excretory structures of flatworms are:

(a) Metanephridia
(b) Malpighian tubules
(c) Flame cells
(d) Nephridiopores only

50. Roundworms belong to phylum:

(a) Annelida
(b) Arthropoda
(c) Platyhelminthes
(d) Nematoda

51. Nematodes are pseudocoelomate because they have:

(a) A false body cavity
(b) No body cavity
(c) A reduced hemocoel
(d) A true coelom lined by mesoderm

52. Which phylum has a rasping organ called radula?

(a) Annelida
(b) Arthropoda
(c) Mollusca
(d) Echinodermata

53. The mantle in molluscs:

(a) Is a respiratory pigment
(b) Secretes a calcareous shell
(c) Forms the radula
(d) Is a larval stage

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54. Metameric segmentation is characteristic of:

(a) Annelida
(b) Nematoda
(c) Cnidaria
(d) Mollusca

55. Setae are absent in:

(a) All annelids
(b) Nereis
(c) Leeches
(d) Earthworms

56. The excretory organs of annelids are:

(a) Metanephridia
(b) Renette cells
(c) Malpighian tubules
(d) Flame cells

57. The most successful invertebrate phylum is:

(a) Annelida
(b) Arthropoda
(c) Mollusca
(d) Echinodermata

58. The excretory organs of insects are:

(a) Metanephridia
(b) Flame cells
(c) Nephridia
(d) Malpighian tubules

59. Shedding of the exoskeleton in arthropods is called:

(a) Ecdysis
(b) Metamerism
(c) Tagmosis
(d) Ossification

60. The body regions of insects formed by fusion of segments are called:

(a) Ossicles
(b) Parapodia
(c) Tagmata
(d) Setae

61. Echinoderms are exclusively:

(a) Terrestrial
(b) Freshwater
(c) Parasitic
(d) Marine

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62. The water-vascular system of echinoderms opens outside through the:

(a) Spiracle
(b) Madreporite
(c) Nephridiopore
(d) Osculum

63. Which of the following is a unique feature of chordates?

(a) Nematocyst
(b) Malpighian tubules
(c) Radula
(d) Notochord

64. In vertebrates, the notochord is replaced by the:

(a) Sternum
(b) Skull
(c) Rib cage
(d) Vertebral column

65. Sea squirts belong to subphylum:

(a) Hemichordata
(b) Urochordata
(c) Vertebrata
(d) Cephalochordata

66. Amphioxus is a member of:

(a) Vertebrata
(b) Urochordata
(c) Cephalochordata
(d) Cyclostomata

67. Fishes have a heart with:

(a) One chamber
(b) Two chambers
(c) Four chambers
(d) Three chambers

68. Amphibians have a heart with:

(a) Two chambers
(b) Four chambers
(c) Three chambers
(d) Five chambers

69. The first group to possess amniotic eggs is:

(a) Aves
(b) Reptilia
(c) Amphibia
(d) Mammalia

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70. Birds are:

(a) Cold-blooded
(b) Ectothermic poikilotherms
(c) Acoelomates
(d) Endothermic homeotherms

71. The voice box of birds is called:

(a) Larynx
(b) Pharynx
(c) Trachea
(d) Syrinx

72. Mammals that lay eggs are called:

(a) Marsupials
(b) Placentals
(c) Monotremes
(d) Primates

73. Kangaroo is a:

(a) Marsupial
(b) Monotreme
(c) Placental
(d) Primate

74. Viruses are not placed in any domain because they are:

(a) Prokaryotic
(b) Eukaryotic
(c) Acellular
(d) Multicellular

75. The protein coat of a virus is called:

(a) Capsomere
(b) Capsid
(c) Envelope
(d) Nucleoid

76. Viruses that infect bacteria are called:

(a) Viroids
(b) Bacteriophages
(c) Prions
(d) Retroviruses

77. HIV is a:

(a) Prion
(b) Bacteriophage
(c) Retrovirus
(d) Viroid

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78. Viroids consist of:

(a) Circular RNA only
(b) RNA and protein
(c) DNA and protein
(d) Protein only

79. Biodiversity is assessed at:

(a) Genetic, species and ecosystem levels
(b) Only genetic level
(c) Only species level
(d) Only ecosystem level

80. Random sampling is important because it:

(a) Increases bias
(b) Requires no identification
(c) Minimizes bias
(d) Eliminates all species

81. A niche is best described as:

(a) Its physical appearance
(b) The role of an organism in its ecosystem
(c) Its trophic level only
(d) The place where it lives only

82. Speciation caused by geographic isolation is:

(a) Sympatric
(b) Parapatric
(c) Peripatric
(d) Allopatric

83. Speciation within the same geographic area is:

(a) Allopatric
(b) Peripatric
(c) Parapatric
(d) Sympatric

84. Darwin's finches are an example of:

(a) Genetic drift only
(b) Allopatric speciation
(c) Sympatric speciation
(d) Artificial selection

85. Polyploidy commonly leads to:

(a) Extinction
(b) Sympatric speciation in plants
(c) Allopatric speciation in animals
(d) Photosynthesis

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ANSWER KEY WITH BRIEF REASONS

1. (d) Robert Whittaker — Whittaker proposed the five-kingdom system in 1969.

2. (d) Carl Woese — Carl Woese separated prokaryotes into Archaea and Bacteria and proposed three domains.

3. (b) Monera — The three domains are Archaea, Bacteria and Eukarya; Monera is a kingdom in the older system.

4. (c) Monera — Kingdom Monera included all prokaryotes.

5. (a) Ether linkages and branched chains — Archaeal membranes have ether-linked branched lipids, unlike ester-linked unbranched lipids of Bacteria/Eukarya.

6. (d) Peptidoglycan — Archaeal walls lack peptidoglycan; they may contain other polysaccharides/proteins or pseudopeptidoglycan.

7. (c) Methanogenesis — Methanogenesis is found in Archaea but not Bacteria or Eukarya.

8. (b) Extremely saline environments — Halobacteria inhabit highly saline environments.

9. (a) Hot environments — Thermococci are associated with hot environments.

10. (c) Carry out anaerobic digestion and produce biogas — Methanogens digest organic matter anaerobically and release methane/biogas.

11. (b) A single circular chromosome — Bacteria have a single circular DNA in the nucleoid without histones.

12. (a) 70S — Bacterial ribosomes sediment at 70S (50S + 30S).

13. (d) Plasmids — Plasmids are self-replicating extra-chromosomal DNA.

14. (a) Bacilli — Bacilli are rod-shaped; cocci spherical, spirilla spiral, vibrios comma-shaped.

15. (d) Vibrios — Vibrios have a comma shape.

16. (b) Streptococci — Streptococci form chains; staphylococci clusters; diplococci pairs.

17. (b) Pili — Pili are involved in attachment and conjugation.

18. (d) Domain — Domain is above kingdom in the hierarchy.

19. (d) Species — Species is the basic/most specific rank.

44. (a) Nematocyst — Nematocysts occur inside cnidocytes and capture prey/defend.

45. (a) Medusa — Medusae are umbrella-like, motile and sexual.

46. (c) Calcium carbonate — Corals produce calcium carbonate exoskeletons.

47. (d) Platyhelminthes — Flatworms are Platyhelminthes.

48. (c) They lack a body cavity — Acoelomates lack a coelom; parenchyma fills the space.

49. (c) Flame cells — Protonephridia with flame cells occur in flatworms.

50. (d) Nematoda — Roundworms are Nematoda.

51. (a) A false body cavity — Nematodes possess a pseudocoelom.

52. (c) Mollusca — Radula is a feeding organ of most molluscs.

53. (b) Secretes a calcareous shell — The mantle surrounds the visceral mass and secretes shell.

54. (a) Annelida — Annelids show true metameric segmentation.

55. (c) Leeches — Leeches lack setae.

56. (a) Metanephridia — Annelids have segmental metanephridia.

57. (b) Arthropoda — Arthropods occupy nearly all habitats and are extremely diverse.

58. (d) Malpighian tubules — Insects excrete via Malpighian tubules.

59. (a) Ecdysis — Ecdysis/molting allows growth.

60. (c) Tagmata — Head, thorax and abdomen are tagmata.

61. (d) Marine — Echinoderms occur only in marine habitats.

62. (b) Madreporite — Madreporite is the sieve-like entry to the water-vascular system.

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20. (c) Order — Sequence: class → order → family → genus → species.

21. (b) Primates — Humans are placed in order Primates.

22. (d) Hominidae — Humans belong to family Hominidae.

23. (a) Angiosperms — The table lists onion under Angiosperms.

24. (d) Chitin — Fungal walls contain chitin.

25. (d) Absorption — Fungi are absorptive heterotrophs.

26. (a) Protozoa — Protozoa are animal-like protists.

27. (a) Have chlorophyll and are autotrophic — Algae contain chlorophyll and photosynthesize.

28. (a) Fungi-like protists — Myxomycota and Oomycota are fungi-like protists.

29. (a) Plasmodium — Plasmodium causes malaria.

30. (b) Trypanosoma — Trypanosoma causes sleeping sickness.

31. (a) Ascomycota — Ascomycota is the largest fungal group.

32. (c) Zygomycota — Rhizopus is a zygomycete with non-septate hyphae.

33. (c) Basidiomycota — Basidiomycota includes mushrooms, rusts and smuts.

34. (a) Penicillium — Penicillin is derived from the fungus Penicillium.

35. (a) Bryophytes — Bryophytes lack xylem and phloem.

36. (b) Naked seeds — Gymnosperms bear naked seeds, usually in cones.

37. (a) Angiosperms — Angiosperms are the most diverse plant group.

38. (b) Multicellular ingestive heterotrophs — Animals ingest food and digest it internally.

39. (b) Porifera — Sponges are Porifera.

40. (b) Ostia — Ostia are incurrent pores; osculum is the excurrent opening.

41. (c) Choanocytes — Choanocytes line the spongocoel and create water currents.

63. (d) Notochord — Notochord is one of the four chordate hallmarks.

64. (d) Vertebral column — The vertebral column replaces the embryonic notochord.

65. (b) Urochordata — Urochordates have notochord only in the larval tail.

66. (c) Cephalochordata — Amphioxus retains the notochord throughout life.

67. (b) Two chambers — Pisces have one auricle and one ventricle.

68. (c) Three chambers — Amphibians have two atria and one ventricle.

69. (b) Reptilia — Reptiles were the first amniotes.

70. (d) Endothermic homeotherms — Birds generate internal heat and maintain body temperature.

71. (d) Syrinx — Syrinx is the avian sound-producing organ.

72. (c) Monotremes — Monotremes such as platypus lay eggs.

73. (a) Marsupial — Kangaroos develop immature young in a pouch.

74. (c) Acellular — Viruses lack cellular organization.

75. (b) Capsid — Capsid surrounds the viral genome.

76. (b) Bacteriophages — Bacteriophages infect bacteria.

77. (c) Retrovirus — HIV contains RNA and reverse transcriptase.

78. (a) Circular RNA only — Viroids are naked circular RNA molecules.

79. (a) Genetic, species and ecosystem levels — Biodiversity includes genetic, species and ecosystem diversity.

80. (c) Minimizes bias — Random sampling gives each part an equal chance and reduces bias.

81. (b) The role of an organism in its ecosystem — Niche includes habitat, interactions and functional role.

82. (d) Allopatric — Allopatric speciation occurs when populations are geographically separated.

83. (d) Sympatric — Sympatric speciation occurs without physical barriers.

84. (b) Allopatric speciation — Island isolation led to allopatric divergence.

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42. (a) Gemmules — Gemmules contain amoeboid cells and help survive harsh conditions.

43. (a) Diploblastic — Cnidarians have two germ layers.

85. (b) Sympatric speciation in plants — Polyploidy can instantly reproductively isolate plant lineages.

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